

SECI Leads with Innovation in India's Clean Energy Mission

Chairman & Managing Director of the Solar Energy Corporation of India Limited (SECI), Mr. R.P. Gupta, is a former IAS officer, has played a vital role in the 'Green goals' of the corporation with his expertise and extensive experience in areas of energy, environment, climate change, de-carbonization strategies, carbon market & trading, sustainability, resource efficiency and material recycling, forest and wildlife. An IITian, Mr. Gupta, is focused on harnessing the vast natural potential for generating clean energy with a visionary zeal.



Mr. R.P. Gupta

Chairman & Managing Director of the Solar Energy Corporation of India Limited (SECI)

Could you share about the genesis of SECI and the 'Green India' vision and mission of the corporation to meet India's increasing energy demands and ensuring energy security?

Solar Energy Corporation of India Limited (SECI) was set up in 2011 to facilitate the implementation of National Solar Mission (NSM) and to achieve the targets set forth by it. SECI was initially incorporated as a Not-for-profit company and later in 2015 converted into a commercial company and its scope was widened to include all of RE resources.

SECI is a Schedule-A CPSU and a Miniratna (Category I) company.

SECI's vision is to be the global driving force in energy transformation towards a greener planet, empowering people and organizations to achieve energy security.

To achieve this vision the company is on a mission-

- To pioneer energy transformation by developing and deploying accessible, reliable and scalable renewable energy solutions.
- To innovate, harness, commercialise and accelerate the use of renewable energy across India, including the remotest corners.
- To be the most trusted partner for all stakeholders and beneficiaries, creating long-term value through responsible and sustainable practices.

The Green India Mission fortunately gels very well with energy security for the country. We are currently dependent on other countries for

import of oil & gas (India imports almost 85% of oil requirement & 50% of its gas requirement). The Green India Mission which uses the locally available natural resources of Sun & Wind, of which India is fortunately abundantly endowed, would help reduce our dependence on energy import by gradually greening the transportation and other sectors which use oil & gas either as fuel or raw material.

Under the Global Renewables and Energy Efficiency Pledge," countries are launching initiatives to achieve alternate, renewable and green energy efficiency for reducing carbon footprint. How does it make the role of SECI significant in enhancing India's capabilities and role in the global context?

SECI's primary mandate is to create an ecosystem that is conducive to the transition of India's energy sector towards renewable energy, which attracts private sector investments. In its pursuit, SECI not only **drives economic growth** but also **cultivates opportunities for employment and fosters sustainable development**.

SECI has awarded **over 65 GW of RE project capacity** through competitive bidding. With **over 21 GW of operational capacity** and numerous projects progressing steadily towards implementation.

In the role of a power procuring intermediary, SECI, by providing **bankability of the agreements**, offers payment security to developers and undertakes to collect payment from DISCOMs. As a result, RE developers are encouraged to offer more **competitive pricing for their electricity**. This ultimately benefits consumers by **making clean energy more affordable and accessible**.

SECI's role extends to encompass pioneering approaches and solutions tailored to India's dynamic energy landscape-

- To scale up green hydrogen production **SECI has awarded projects for generation of 412,000 metric tonnes of green hydrogen and electrolyser manufacturing capacity of 1.5 GW.**
- SECI has introduced pioneering approaches such as **Hybrid, Round-the-Clock (RTC), and Firm and Dispatchable RE (FDRE) models**. These

models ensure a more reliable and stable supply of renewable energy, addressing concerns related to intermittency.

- SECI collaborates closely with relevant stakeholders to ensure the success of its initiatives. By **engaging with government agencies, industry partners, and local communities**, SECI fosters collaboration and knowledge sharing, ultimately enhancing the impact of RE.

India, is in the elite top five economies of the world and is aiming to move up to the top three in the next couple of years. How do you see SECI contributing towards meeting this ambitious target?

Currently India is **the 5th largest with a GDP of \$3.7 trillion** (estimate FY24), and India is expected to become the **third-largest economy in the world with a GDP of \$5 trillion in the next three years and touch \$7 trillion by 2030**. Electricity is the most important infrastructure which is a sine qua non for development.

Peak power demand in India is likely to cross 400 GW by 2030 and more generation capacity is being set up to meet the rising demand, **The demand grew at 9 per cent in 2023, and is growing at 10 per cent in 2024.**

To achieve the target of 500 GW by 2030, Government has declared plan to **add 50 GW of RE capacity annually for next 5 years (until FY 28)**. SECI has been issuing tenders and has awarded over 65 GW of RE capacity.

In response to the growing share of RE and to **better integrate the RE capacities into the National Grid**, the company has been at the forefront of innovation, introducing pioneering approaches such as **Hybrid, Round-the-Clock (RTC), and Firm and Dispatchable RE (FDRE) models**.

India is gifted with abundant sunshine making solar energy an ideal source of clean and renewable energy. How is SECI harnessing this limitless natural potential by emphasizing on use of advance technology for over-all efficiency, reducing costs from manufacturing to maintenance and durability under stressful climatic conditions?

India is endowed with vast solar energy potential. About **5,000 trillion kWh per year energy**